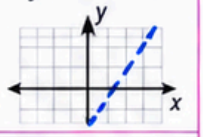
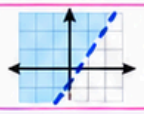
Algebra Graphing Inequalities on the Coordinate Plane Cheat Sheet

Unit Focus: *graphing and interpreting linear inequalities*

1 KEY VOCABULARY

inequality	A statement that compares two expressions using $<$, $>$, $\leq$, or $\geq$.	$y > 2x - 1$
boundary line	The dashed or solid line that separates the coordinate plane into two regions. It comes from replacing the inequality with an equation (use $=$).	$y = 2x - 1$ 
solution set	All points (x, y) that make the inequality true.	Shown by the shaded region.
test point	A point not on the boundary line (usually $(0, 0)$) used to determine which side to shade.	$(0, 0)$
shaded region	The side of the boundary line that contains all the solutions.	

2 INEQUALITY SYMBOLS

Each symbol shows how two expressions are compared.

Symbol	Meaning	Graph Feature	Example
$>$	greater than	dashed line	$y > 2x + 1$
$<$	less than	dashed line	$y < -x + 4$
$\geq$	greater than or equal to	solid line	$y \geq x - 3$
$\leq$	less than or equal to	solid line	$y \leq -2x + 5$

Use a dashed line for $>$ or $<$.
Use a solid line for $\geq$ or $\leq$.

The boundary line is always graphed using an equation (with $=$).

3 REWRITING IN SLOPE-INTERCEPT FORM

It is often easiest to graph a linear inequality in slope-intercept form $y = mx + b$. If needed, solve for y using the same steps as equations.

Example: Rewrite $2x + y \geq 4$ in slope-intercept form.

Step 1: Subtract $2x$ from both sides.

$y \geq -2x + 4$

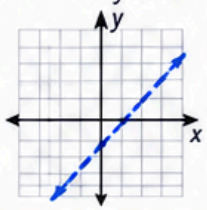
Now graph $y \geq -2x + 4$.

4 SOLID VS. DASHED BOUNDARY LINES

The boundary line comes from the equation (replace the inequality with $=$). Use a solid or dashed line based on the inequality symbol.

Dashed Line ($>$ or $<$)

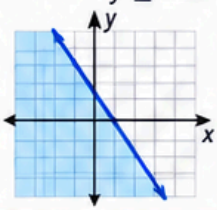
$y > x - 1$



Dashed line because the line itself is not included in the solution set.

Solid Line ($\geq$ or $\leq$)

$y \leq -2x + 3$



Solid line because the line is included in the solution set.

5 GRAPHING A LINEAR INEQUALITY

Follow these steps to graph any linear inequality.

1. Rewrite in slope-intercept form (if needed): $y = mx + b$.
2. Graph the boundary line using a solid or dashed line.
3. Choose a test point (usually $(0, 0)$).
4. Substitute the test point into the inequality.
5. If the test point makes the inequality true, shade that side. If false, shade the other side.

6 USING A TEST POINT

Plug a point (usually $(0, 0)$) into the inequality to decide which side to shade.

Example 1: $y > 2x - 1$

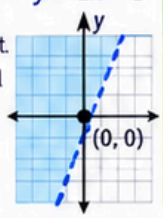
Step 1: Graph the boundary line $y = 2x - 1$ (dashed).

Step 2: Use $(0, 0)$ as a test point.

Step 3: Substitute: $0 > 2(0) - 1$

$0 > -1$ ✓ (true)

Step 4: Shade the side that contains $(0, 0)$.



Example 2: $y \leq -x + 4$

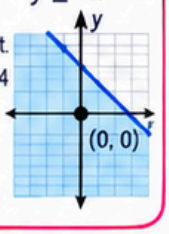
Step 1: Graph the boundary line $y = -x + 4$ (solid).

Step 2: Use $(0, 0)$ as a test point.

Step 3: Substitute: $0 \leq -(0) + 4$

$0 \leq 4$ ✓ (true)

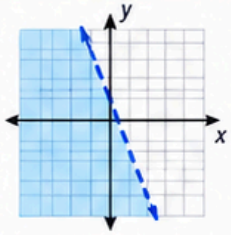
Step 4: Shade the side that contains $(0, 0)$.



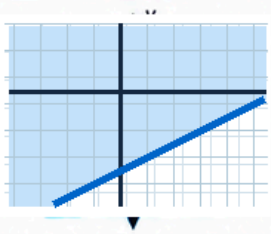
7 MORE EXAMPLES

Graph each linear inequality.

a) $y < -3x + 2$

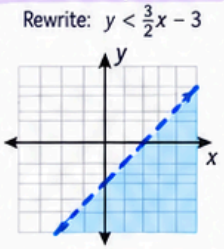


b) $y \geq \frac{1}{2}x - 3$



c) $3x - 2y > 6$

Rewrite: $y < \frac{3}{2}x - 3$



8 QUICK REMINDERS & COMMON MISTAKES

DO REMEMBER:

- Use a solid line for $\geq$ or $\leq$ and a dashed line for $>$ or $<$.
- Use a test point to decide which side to shade.
- The solution set is the shaded region (including the line for $\geq$ or $\leq$).
- Rewrite in slope-intercept form if needed.
- Check your work by testing a point in the shaded region.

COMMON MISTAKES:

- Using a solid line for $>$ or $<$. Forgetting to shade a region.
- Shading the wrong side (because the test point was incorrect).
- Graphing the inequality without first using $=$ for the boundary line.
- Testing a point on the line (instead of a point not on the line).
- Mixing up the inequality symbol when solving for y.

Front

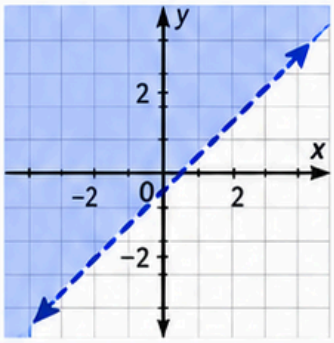
Algebra Graphing Inequalities on the Coordinate Plane Cheat Sheet

More Practice: shading, test points, and systems

1 WRITING AN INEQUALITY FROM A GRAPH

Use the boundary line and shading to write the inequality.

Example: Write an inequality for the graph.



Step 1: Find the boundary line.
$y = x - 1$

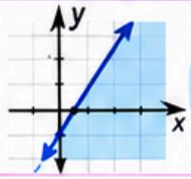
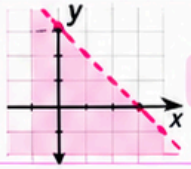
Step 2: Determine ≤ or <.
Dashed line → strict inequality (< or >).

Step 3: Use the shading.
Shaded above the line → $y > x - 1$

Answer: $y > x - 1$

2 BOUNDARY LINE RULES

The boundary line separates the plane into two regions.

Inequality	Boundary Line	Example Graph
≤ or ≥	**Solid line** (points on the line are included)	 $y \geq 2x + 1$
< or >	**Dashed line** (points on the line are not included)	 $y < -x + 2$

3 GRAPHING A LINEAR INEQUALITY

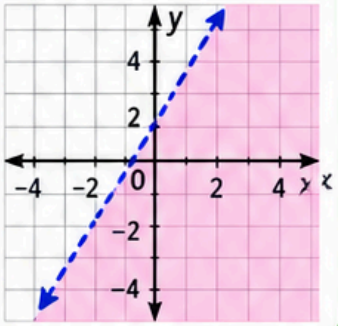
Use a test point to determine which side to shade.

Example: Graph $y < 2x + 3$.

Step 1: Graph the boundary line $y = 2x + 3$ using a dashed line (<).

Step 2: Use a test point, (0, 0).
$0 < 2(0) + 3$
$0 < 3$ ✓ (true)

Step 3: Shade the side that contains the test point.



4 SYSTEMS OF LINEAR INEQUALITIES

Graph each inequality and shade the overlapping region.

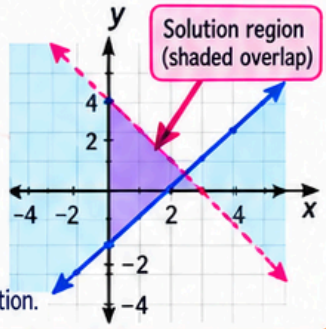
Example: Graph the system.

$y \geq x - 1$
$y < -x + 2$

Step 1: Graph $y \geq x - 1$ (solid line, shade above).

Step 2: Graph $y < -x + 2$ (dashed line, shade below).

Step 3: The overlapping region is the solution.

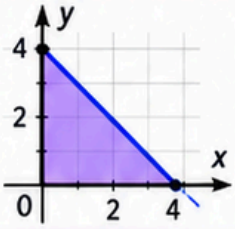


5 FEASIBLE REGION

The feasible region (or solution region) is the set of all points that satisfy all inequalities in a system.

Example: Identify the feasible region.

$x \geq 0$
$y \geq 0$
$x + y \leq 4$



The feasible region is the shaded triangle, including the boundary lines (solid lines).

6 WORKED EXAMPLE: GRAPHING A SYSTEM

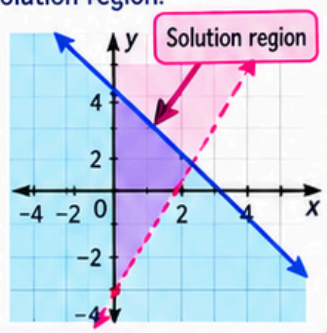
Graph the system and identify the solution region.

Example:
$y \leq -x + 4$
$y > 2x - 2$

Step 1: Graph $y \leq -x + 4$ (solid line, shade below).

Step 2: Graph $y > 2x - 2$ (dashed line, shade above).

Step 3: The overlapping shaded region is the solution.



7 USING A TEST POINT

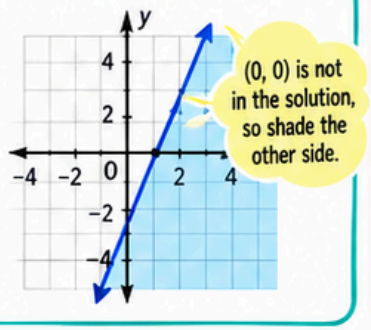
Pick a point not on the boundary line to determine which side to shade. (0, 0) is usually the easiest test point.

Example: Graph $3x - y \geq 6$.

Step 1: Rewrite if needed: $y \leq 3x - 6$

Step 2: Test point (0, 0).
$0 \leq 3(0) - 6$
$0 \leq -6$ ✗ (false)

Step 3: Shade the side that does NOT contain the test point.



8 QUICK REMINDERS & COMMON MISTAKES

✓ DO REMEMBER:

- Use a dashed line for < or > and a solid line for ≤ or ≥.
- Use a test point to decide which side to shade.
- The solution to a system is the overlapping shaded region.
- The feasible region satisfies all inequalities.
- Check if a point is in the solution by substituting it into each inequality.

✗ COMMON MISTAKES:

- Using the wrong type of line (solid vs. dashed).
- Shading the wrong side (check with a test point).
- Forgetting the solution to a system is the overlap, not the union.
- Not including boundary lines for ≤ or ≥.
- Using a point on the boundary line as a test point.